

# Engineering Design Report

Fractional Forge — Project a\_compact\_commercial\_airsource\_heat\_pump

Generated: 2026-05-05

## Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£30.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

### Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction

[E] LLM assumption or unverified hypothesis

# Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

| Check                              | Status | Reason                        | Evidence     |
|------------------------------------|--------|-------------------------------|--------------|
| 1. Market TAM/SAM/SOM Defined      | FAIL   | Missing market data           | Not computed |
| 2. Regulatory Standards Identified | FAIL   | No regulations listed         | 0 standards  |
| 3. Cost Ceiling Specified          | PASS   | Target provided               | £30.00       |
| 4. Spatial Envelope Provided       | PASS   | Volume established            | 75 m³        |
| 5. Modules Decomposed              | PASS   | System architecture built     | 10 modules   |
| 6. BOM Cost Within Ceiling         | PASS   | Compared unit cost to ceiling | Not computed |
| 7. Risk Matrix Saturated           | PASS   | FMEA rows generated           | Risks exist  |

### Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

### Data Sources

[System] Internal gate checks

Overall Section Grade: A

# 1. Brief & Requirements [A]

---

## 1.1 Mission & Use Case

---

Mission

Not computed

Use Case

Not computed

## 1.2 Market Context & Timing [B]

---

Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

# 1.3 Competitor Landscape [C]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Not computed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                           |
| Product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Ecodan CAHV-R450YA-HP (Hydronic commercial range, ~45kW, cascadeable)                                                                                                                     |
| Pricing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Industry sources suggest installed costs for commercial Ecodan systems in the 30–50kW range typically exceed £15,000–£25,000 installed; unit-only list pricing not consistently published |
| Tech Specs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Not computed                                                                                                                                                                              |
| <div><div>STRENGTHS</div><div>Established brand recognition and installer network in UK,Comprehensive MCS and ErP documentation,Strong technical support and warranty infrastructure,Modular design allows capacity scalability</div></div> <div><div>WEAKNESSES</div><div>R32 refrigerant with GWP ~675, subject to F-Gas pressure,Larger form factor than proposed product — may not suit very constrained plant rooms,Premium pricing limits accessibility for cost-sensitive small commercial clients,Not specifically optimised for the 25–35kW stand-alone segment</div></div> |                                                                                                                                                                                           |
| <div>DIFFERENTIATION ANGLE</div> <div>Not computed</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                           |

## Not computed

|            |                                                                                                                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product    | aroTHERM plus VWL 125/5 AS S2 (commercial-adjacent, ~12.6kW; larger aroTHERM commercial range up to ~75kW)                                                                                                                                                      |
| Pricing    | Published trade pricing for aroTHERM commercial units in the 30kW range is not consistently available; industry estimates suggest mid-range European brand commercial units in this class are priced at £6,000–£12,000 unit cost at typical distributor margins |
| Tech Specs | Not computed                                                                                                                                                                                                                                                    |

### STRENGTHS

Vaillant aroTHERM plus residential range demonstrates R290 commercial viability, Strong UK brand presence and installer training programmes, MCS listed products in residential range, Broad product portfolio enabling system selling

### WEAKNESSES

Gap in Vaillant portfolio specifically at 30kW R290 hydronic split for small commercial MCS applications as of 2023–2024, Larger commercial Vaillant units typically use higher-GWP refrigerants, Installer channel skewed toward residential; commercial installer base smaller

### DIFFERENTIATION ANGLE

Not computed

## Not computed

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product    | Daikin Altherma 3 M (monobloc, up to ~16kW residential/light commercial) and Daikin Altherma H HT (high temperature, commercial range)                                                 |
| Pricing    | Industry commentary suggests Daikin commercial heat pump systems in the 30–60kW range carry installed costs substantially above £20,000; unit pricing not consistently publicly listed |
| Tech Specs | Not computed                                                                                                                                                                           |

### STRENGTHS

Dominant global HVAC brand with extensive UK commercial HVAC installer network, Strong ErP and regulatory compliance documentation, Wide after-sales support infrastructure, Established commercial projects track record

### WEAKNESSES

R32 and R410A refrigerants face increasing F-Gas regulatory headwinds, Product line not specifically targeted at the sub-50kW standalone small commercial hydronic segment with R290, Premium brand pricing, Complexity of commercial product range can create specification friction

### DIFFERENTIATION ANGLE

Not computed

## Not computed

|            |                                                                                                                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product    | WPL series (air-to-water heat pumps, including WPL 47 HK and WPL 57 HK for larger outputs)                                                                                                                              |
| Pricing    | Published and trade sources suggest Stiebel Eltron WPL commercial-range units are priced in the £8,000–£15,000 unit range depending on output and specification; precise list pricing requires distributor verification |
| Tech Specs | Not computed                                                                                                                                                                                                            |

### STRENGTHS

German engineering reputation with strong reliability perception, Established UK distribution through specialist heating merchants, Hydronic architecture aligns with proposed product, Comprehensive documentation for MCS/ErP compliance processes

### WEAKNESSES

R410A on several core commercial models, increasing regulatory exposure, Premium pricing relative to the proposed target cost, Smaller UK installer network compared to Mitsubishi Electric or Daikin, Less aggressive UK commercial marketing presence

### DIFFERENTIATION ANGLE

Not computed

## Not computed

|            |                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product    | Thermia Mega / Thermia Atlas commercial heat pumps (ground and air source commercial range, typically 30–150kW)                                                                                                                           |
| Pricing    | Industry sources suggest Thermia commercial air-source units are positioned as premium/specification-grade products; indicative pricing in the 30kW class may exceed £10,000 unit cost, but publicly available UK list pricing is limited |
| Tech Specs | Not computed                                                                                                                                                                                                                              |

### STRENGTHS

Purpose-designed for commercial heating loads in the 30kW+ range, Hydronic architecture directly comparable to proposed product, Danfoss group ownership provides component supply security and engineering depth, Strong reference base in Scandinavian and Northern European climates relevant to UK conditions

### WEAKNESSES

Limited brand recognition among UK M&E contractors and small commercial specifiers, R410A exposure on legacy models, Distribution network in UK is thinner than tier-1 HVAC brands, Pricing and positioning not optimised for cost-sensitive small commercial UK buyers

### DIFFERENTIATION ANGLE

Not computed

### Data Sources

[LLM Search] Aggregated competitor data via LLM

**Overall Section Grade: D**



# 1.4 Constraints & Targets <sup>[A]</sup>

|                     |              |
|---------------------|--------------|
| Target Material     | Not computed |
| Target Process      | Not computed |
| Tolerance Target    | Not computed |
| Quantity Target     | Not computed |
| Batch Size          | 500          |
| Target Cost Ceiling | £30.00       |
| Max Mass            | Not computed |

**Data Sources**

[Mixed] Constraints provided by User, Sources generated by LLM

**Overall Section Grade: B**

## 2. Regulatory & Compliance

---

No regulatory standards identified for this brief.

### 3. Sizing & Spatial Optimisation <sup>[D]</sup>

#### INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

#### System Envelope

|                             |                       |
|-----------------------------|-----------------------|
| Internal Dimensions (W×D×H) | 5000 × 5000 × 3000 mm |
| Internal Floor Area         | 25 m <sup>2</sup>     |
| Internal Volume             | 75 m <sup>3</sup>     |
| Rules Domain                | hardware_product      |

#### Module Allocation Zone Table

| Module                                        | L × W × H (mm)     | Area (m <sup>2</sup> ) | Mount |
|-----------------------------------------------|--------------------|------------------------|-------|
| outdoor_structural_chassis_enclosure          | 2236 × 2236 × 2400 | 5                      | floor |
| evaporator_airward_heat_capture_system        | 2236 × 2236 × 2400 | 5                      | floor |
| r290_inverter_compression_node                | 2236 × 2236 × 2400 | 5                      | floor |
| dynamic_refrigerant_flow_reversing_control    | 2236 × 2236 × 2400 | 5                      | floor |
| outdoor_primary_thermo_hydraulic_exchanger    | 2236 × 2236 × 2400 | 5                      | floor |
| ex_proof_r290_leak_mitigation_control_node    | 2236 × 2236 × 2400 | 5                      | floor |
| outdoor_power_electronics_inverter_drive      | 2236 × 2236 × 2400 | 5                      | floor |
| indoor_hydraulic_circulation_pressure_control | 2236 × 2236 × 2400 | 5                      | floor |
|                                               |                    |                        |       |

|                                            |                    |   |       |
|--------------------------------------------|--------------------|---|-------|
| indoor_peak_load_emergency_auxiliary_block | 2236 × 2236 × 2400 | 5 | floor |
| system_telemetry_master_control_panel      | 2236 × 2236 × 2400 | 5 | floor |

**Conflicts**

- Total module footprint (50.0m2) exceeds envelope budget (21.3m2)

**Data Sources**

[Deterministic Solver] 3D Box Packing Constraints Engine

**Overall Section Grade: B**

# 4.1 Outdoor Structural Chassis & Enclosure [D]

Maturity: draft

Purpose

Provides weatherproofing, structural rigidity, structural isolation for the sealed refrigeration cycle, and acoustic damping to meet planning limits.

Why It Matters

Not computed

Technical Description

The outer chassis forms the structural boundary of the 30 kW heat pump unit. All primary frame members and panels are fabricated from 2 mm minimum hot-dip galvanised steel to BS EN ISO 1461 . External panel surfaces receive a thermosetting powder-coat finish rated to corrosivity category C4/C5 per ISO 12944, providing a service life of at least 15 years in coastal or industrial environments .

The internal faces of the compressor bay panels are lined with 30 mm closed-cell EPDM acoustic absorption foam. This lining targets attenuation of mid-frequency noise in the 500 Hz to 2 kHz octave bands, contributing to the design sound power ceiling of 68 dB(A) at 1 m . The compressor sub-base is mechanically decoupled from the main chassis via heavy-duty elastomeric anti-vibration mounts rated at ±15 mm static deflection . These mounts reduce transmission of compressor-generated vibration below 10 Hz into the concrete plinth . The electrical bay and the R290 compressor bay are physically segregated by a 1.5 mm steel partition, isolating potential ignition sources from the refrigerant circuit in compliance with EN 60079-10 zoning principles .

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

## 4.2 Evaporator & Airward Heat Capture System [D]

Maturity: draft

### Purpose

Extracts low-grade thermal energy from the ambient air down to -20°C by facilitating the phase change of the R290 refrigerant.

### Why It Matters

Not computed

### Technical Description

The evaporator is a multi-row fin-and-tube heat exchanger spanning the full 1,200 mm rear profile of the outdoor unit . The coil uses 3/8-inch outer diameter copper tubes internally grooved with 60 helical ridges to enhance refrigerant-side heat transfer by approximately 20% versus smooth-bore equivalents . Aluminium fins are mechanically expanded onto the tubes at a pitch of 1.8 mm and stamped with a louvred profile to induce local flow turbulence. The fin surfaces receive a 5 µm hydrophilic epoxy resin coating, reducing water contact angle below 20° to accelerate condensate run-off and delay ice bridge formation at ambient temperatures between 0°C and -5°C .

Two 500 mm diameter axial fans, vertically stacked, deliver a combined rated airflow of 6,500 m³/h at maximum speed . The fans are driven by electronically commutated (EC) motors with nominal efficiency of 75% at full load . Motor speed modulates from 300 RPM to 950 RPM in response to ambient temperature and capacity demand signals from the master controller. A 304 stainless steel drip tray, sloped at 3° towards the 32 mm drain outlet, sits beneath the coil . A 40 W PTC silicone-insulated trace heater maintains the tray above 4°C during reverse-cycle defrost intervals lasting 30 to 90 minutes .

### Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**



# 4.3 R290 Inverter Compression Node [D]

Maturity: draft

Purpose

Elevates the pressure and temperature of the R290 vapour, driving the thermodynamic heating cycle.

Why It Matters

Not computed

Technical Description

The compression node contains an ATEX-rated variable-speed scroll compressor optimised for R290 (propane, GWP = 3) as the working fluid . The compressor displaces approximately 35 cm³ per revolution at 50 Hz and is lubricated with a Polyolester (POE) refrigeration oil of ISO VG 32 viscosity grade . This POE formulation is specifically compounded to resist viscosity reduction below 5 cSt when exposed to R290's high solubility in oil at crankcase temperatures below 20°C . A 45 W silicone rubber crankcase heater maintains sump oil temperature at least 15 K above ambient during off-cycles, preventing liquid R290 migration into the oil volume at ambient temperatures down to -20°C .

The inverter drive modulates compressor speed from 20 Hz to 90 Hz, corresponding to approximately 25% to 100% of the rated 30 kW heating capacity . This range eliminates short-cycling below 6-minute run periods that would otherwise cause excessive oil return failure. A 50 mm thick multi-layer acoustic and thermal jacket, combining 25 mm mass-loaded vinyl and 25 mm closed-cell foam, wraps the compressor shell and reduces radiated airborne noise by up to 8 dB(A) at 1 m . A discharge-line pulsation muffler with a 0.5-litre internal volume attenuates gas-borne pressure pulses in the 50 to 200 Hz range .

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.4 Dynamic Refrigerant Flow & Reversing Control [D]

Maturity: draft

## Purpose

Accurately throttles high-pressure liquid refrigerant to low-pressure vapour and logically switches the circuit between heating and active defrost.

## Why It Matters

Not computed

## Technical Description

Refrigerant flow is controlled primarily by a 500-step unipolar stepper motor Electronic Expansion Valve (EEV) with a minimum orifice resolution of 0.06 mm<sup>2</sup> per step . The master controller executes a proportional-integral superheat control loop at 10-second intervals, targeting a suction superheat of 3 K to 5 K at the compressor inlet . Step position adjustments occur up to 360 times per hour under fluctuating ambient conditions, preventing both liquid slugging and excessive superheat losses .

Flow direction is governed by a pilot-operated brass 4-way reversing valve with a Cv of 4.2 . During heating operation, the valve routes high-pressure discharge gas to the indoor brazed-plate condenser. A 24 V DC solenoid pilot energises to reverse flow direction within 0.5 seconds, directing hot discharge gas into the outdoor evaporator coil to initiate defrost . Defrost cycles activate when the coil NTC thermistor reads below -8°C and terminate when coil temperature recovers above 12°C . A solid-core bi-flow filter drier containing 80 g of molecular sieve desiccant protects the EEV orifice from particulate matter larger than 25 µm and maintains system moisture content below 10 ppm .

## Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.5 Outdoor Primary Thermo-Hydraulic Exchanger [D]

Maturity: draft

Purpose

Transfers heat from the hot, compressed R290 gas into the sealed indoor hydronic water/glycol loop.

Why It Matters

Not computed

Technical Description

Heat transfer between the R290 refrigerant circuit and the hydronic water/glycol loop is accomplished by a Braze Plate Heat Exchanger (BPHE) rated at 32 kW thermal duty at standard conditions of 35°C inlet water and 45°C outlet . The BPHE consists of 40 alternating 316L stainless steel plates, each 0.4 mm thick, stamp-formed with a 60° herringbone corrugation pattern and vacuum-brazed with 99.9% pure copper at 1,083°C . Counter-flow arrangement between the refrigerant and water/glycol streams achieves a log mean temperature difference (LMTD) of 4 K to 6 K at full load . The asymmetric plate geometry maximises condensing-side heat transfer while keeping the water-side pressure drop below 25 kPa at 1.5 m³/h flow .

Confining the BPHE within the outdoor enclosure ensures zero R290 penetration into the building fabric, directly satisfying EN 378-3:2016 clause 6.3 requirements for A2L and A3 refrigerants in occupied buildings . A 20 mm thick moulded closed-cell polyurethane foam jacket, vapour-sealed with an aluminium foil laminate, maintains the BPHE surface temperature above the dew point during sub-zero ambient operation .

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.6 Ex-Proof R290 Leak Mitigation & Control Node <sup>[D]</sup>

Maturity: draft

Purpose

Continuously monitors the sealed outdoor chassis for refrigerant leaks and triggers active electrical isolation and purging.

Why It Matters

Not computed

Technical Description

An infrared (IR) non-dispersive combustible gas sensor is mounted 50 mm above the compressor bay floor, exploiting propane's vapour density of 1.52 relative to air to detect accumulations at the lowest chassis point . The sensor detects R290 concentrations from 100 ppm up to the Lower Explosive Limit (LEL) of 21,000 ppm (2.1% v/v in air), with a calibrated alarm threshold set at 15% LEL (3,150 ppm) . The sensor response time T90 is less than 15 seconds under stagnant air conditions .

On alarm, the safety microprocessor commands two solid-state hermetic isolation relays to de-energise the compressor inverter drive and all non-intrinsically safe panel loads within 200 ms . Simultaneously, an ATEX Group IIA, Category 2G certified DC centrifugal purge blower rated at 120 m³/h activates . At this flow rate, the 0.35 m³ compressor bay volume undergoes a minimum of 5 air changes per minute, diluting a worst-case 500 g R290 leak below 15% LEL within 90 seconds . A concurrent critical fault frame is transmitted over the Modbus RTU link to the building BMS within 3 communication cycles .

Bill of Materials (BOM) <sup>[D]</sup>

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**



# 4.7 Outdoor Power Electronics & Inverter Drive [D]

Maturity: draft

Purpose

Rectifies grid AC to DC, then synthesises variable-frequency AC power for the compressor, whilst actively filtering harmonic distortion.

Why It Matters

Not computed

Technical Description

The inverter drive accepts 400 V, 3-phase, 50 Hz grid input at a maximum continuous current of 52 A and converts it to variable-frequency AC for the scroll compressor . Incoming supply passes through a 3-phase LC line choke of 3 mH inductance per phase, attenuating conducted harmonic emissions to comply with BS EN 61000-3-12 current harmonic limits for equipment above 16 A . A three-phase bridge rectifier feeds a 680 µF, 800 V rated aluminium electrolytic DC link capacitor bank, maintaining DC bus ripple below 5% at full load .

The switching stage uses an Insulated-Gate Bipolar Transistor (IGBT) Intelligent Power Module (IPM) rated at 600 V, 75 A with a switching frequency of 8 kHz . A dedicated DSP running at 150 MHz executes the sinusoidal pulse-width modulation (SPWM) algorithm, producing a 20 Hz to 90 Hz output waveform with total harmonic distortion below 3% . The IPM die is bonded directly to a 200 mm × 150 mm aluminium cold plate through which cool liquid R290 circulates via a 6 mm diameter bypass capillary at approximately 0.8 kg/min . This arrangement limits IPM junction temperature to a maximum of 65°C at 30 kW peak output during ambient conditions of 35°C .

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

## 4.8 Indoor Hydronic Circulation & Pressure Control <sup>[D]</sup>

Maturity: draft

### Purpose

Manages system pressure, expansion, and highly controlled water/glycol mixture circulation to the commercial heat emitters.

### Why It Matters

Not computed

### Technical Description

The indoor hydrobox circulates heated water/glycol through the building distribution system using a DN32 glandless EC circulator pump with a maximum head of 10 m at 3.0 m<sup>3</sup>/h . The permanent magnet rotor motor draws a maximum of 185 W at full speed and is modulated via a 0–10 V PWM signal from the master control board . The control loop maintains a constant differential temperature of 5 K across the heat emitter circuit at design flow temperatures of 35°C to 55°C .

Volumetric thermal expansion of 45 litres of 25% propylene glycol solution heating from 10°C to 55°C generates approximately 1.1 litres of expansion fluid . A 12-litre EPDM diaphragm expansion vessel with a 1.5 bar pre-charge pressure absorbs this expansion, maintaining system pressure between 1.0 bar and 2.5 bar at all operating points . A 3-bar brass pressure relief valve with a 22 mm discharge pipe routed to an external safe drain provides overpressure protection independent of all control systems . A 32 mm inline magnetic microbubble dirt separator captures ferrous magnetite particles down to 30 µm, preventing particulate transport to the 0.4 mm BPHE plates in the outdoor unit .

### Bill of Materials (BOM) <sup>[D]</sup>

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.9 Indoor Peak-Load & Emergency Auxiliary Block <sup>[D]</sup>

Maturity: draft

Purpose

Provides direct electrical resistance heating to cover peak commercial demands during deep freeze snaps or emergency heat if the compressor faults.

Why It Matters

Not computed

Technical Description

The auxiliary heating block provides up to 9 kW of direct electrical resistance heating via a 3-phase Incoloy 800 immersion element housed in a 304 stainless steel flow-through pressure vessel rated to 6 bar working pressure . The Incoloy 800 alloy sheath operates at a maximum watt density of 3.5 W/cm² to limit fouling deposits on the element surface in hard-water conditions above 150 mg/L calcium carbonate .

The 9 kW total load is staged in three 3 kW increments, with each increment switched by a dedicated silent-switching solid-state contactor rated at 25 A continuous . The control board activates Stage 1 only when the outdoor ambient drops below the -7°C bivalent point or when the heat pump coefficient of performance falls below 1.8 . All 3 stages activate during 65°C legionella pasteurisation cycles, which run for a minimum of 60 minutes per EN 806-2 guidance . Regardless of PLC commands, two independent mechanical high-limit thermostats strapped 50 mm apart on the vessel body sever the 240 V control circuit at 85°C, protecting against dry-fire conditions .

Bill of Materials (BOM) <sup>[D]</sup>

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**

# 4.10 System Telemetry & Master Control Panel [D]

Maturity: draft

Purpose

Orchestrates the entire system logic, dynamically processing external weather data and facility demands to manipulate compressor/pump speeds.

Why It Matters

Not computed

Technical Description

The master control panel is built around an industrial ARM Cortex-M7 processor running at 216 MHz, executing a real-time operating system (RTOS) with a 1 ms control loop cycle time . The system continuously acquires data from 22 discrete field instruments, comprising NTC thermistors, 4–20 mA pressure transducers, and paddle-type flow meters, via a 16-channel 24-bit analogue-to-digital converter . The primary control algorithm implements a weather compensation curve, calculating the required flow temperature from the measured outdoor ambient between -20°C and +20°C with a resolution of 0.1 K, to maintain maximum seasonal COP .

The board hosts two RS485 physical layer transceivers supporting both Modbus RTU at 19,200 baud and Modbus TCP/IP via an integrated 100 Mbit Ethernet port, enabling integration into building management systems at up to 247 node addresses . Operating personnel interact with the system through a front-mounted 5-inch, 800 × 480 pixel capacitive touchscreen HMI displaying a live mimic diagram of the vapour compression cycle, 12-month fault log storage of up to 500 events, and password-protected commissioning override menus .

Bill of Materials (BOM) [D]

| Part #            | Description | Supplier | Grade | Cost |
|-------------------|-------------|----------|-------|------|
| No parts defined. |             |          |       |      |

### Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

**Overall Section Grade: D**



# 5. Cost Waterfall & Economics [D]

## Unit Cost Breakdown

| Module              | Subtotal     |
|---------------------|--------------|
| Total Unit BOM Cost | Not computed |

## Non-Recurring Engineering (NRE)

| Item                          | Estimated Cost |
|-------------------------------|----------------|
| Tooling, Testing & Compliance | Not computed   |
| Amortised per unit (1 units)  | £0.00          |

## Cost Reduction Paths

| Strategy                              | Est. Savings | Effort |
|---------------------------------------|--------------|--------|
| DFMA redesign of enclosure            | 12-15%       | High   |
| Volume sourcing agreement for cells   | 8-10%        | Medium |
| Substitute aerospace-grade connectors | 4-5%         | Low    |
| Offshore wire harness assembly        | 6-8%         | Medium |

**Data Sources**

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

**Overall Section Grade: C**

## 6. Failure Modes & Effects Analysis (FMEA) <sup>[D]</sup>

| Module                                       | Hazard                                            | Cause        | Effect       | S×L=RPN | Mitigation & Verification Test                                                                     |
|----------------------------------------------|---------------------------------------------------|--------------|--------------|---------|----------------------------------------------------------------------------------------------------|
| Outdoor Structural Chassis & Enclosure       | Corrosion compromising structural integrity       | Not computed | Not computed | 3×2=6   | Utilise marine-grade C5 powder-coating and stainless steel 316 A4 fasteners.                       |
| Outdoor Structural Chassis & Enclosure       | Excessive noise resulting in planning enforcement | Not computed | Not computed | 4×3=12  | Multi-layer acoustic baffling and rigorous final assembly vibration testing.                       |
| Evaporator & Airward Heat Capture System     | Total blockage of airflow via ice accumulation    | Not computed | Not computed | 4×3=12  | Redundant thermistors verifying complete clearing of ice before defrost cycle termination.         |
| Evaporator & Airward Heat Capture System     | EC motor electrical failure                       | Not computed | Not computed | 4×2=8   | Specify sealed IP55 rated motors with integral thermal overload protection.                        |
| R290 Inverter Compression Node               | Compressor seizure due to liquid slugging         | Not computed | Not computed | 5×2=10  | Crankcase heater interlocked with start-up sequence and strict superheat control via EEV.          |
| R290 Inverter Compression Node               | Spark ignition of local R290 leaks                | Not computed | Not computed | 5×1=5   | Hermetically sealed compressor terminals and ATEX certification for internal components.           |
| Dynamic Refrigerant Flow & Reversing Control | System freezing over due to defrost failure       | Not computed | Not computed | 4×3=12  | Firmware logic that triggers aggressive fallback defrost or trips an alarm before total ice-block. |
|                                              |                                                   |              |              |         |                                                                                                    |

|                                                           |                                                   |              |              |               |                                                                                                              |
|-----------------------------------------------------------|---------------------------------------------------|--------------|--------------|---------------|--------------------------------------------------------------------------------------------------------------|
| <b>Dynamic Refrigerant Flow &amp; Reversing Control</b>   | Moisture reacting with POE oil to form sludge     | Not computed | Not computed | <b>3×2=6</b>  | Oversized high-capacity solid core filter drier and strict vacuum manufacturing protocols.                   |
| <b>Outdoor Primary Thermo-Hydraulic Exchanger</b>         | R290 leakage directly into hydronic water piping  | Not computed | Not computed | <b>5×1=5</b>  | PED-compliant BPHE hydrostatic testing at 1.5x design pressure; external safety valves on water side.        |
| <b>Outdoor Primary Thermo-Hydraulic Exchanger</b>         | Water freezing rupturing the heat exchanger       | Not computed | Not computed | <b>4×3=12</b> | Mandatory anti-freeze glycol requirement for the loop and integrated low-temperature pump circulation logic. |
| <b>Ex-Proof R290 Leak Mitigation &amp; Control Node</b>   | Ignition of pooled R290 gas within the chassis    | Not computed | Not computed | <b>5×1=5</b>  | Use of IR instead of catalytic sensors for drift-free long life; weekly self-test of the purge fan.          |
| <b>Ex-Proof R290 Leak Mitigation &amp; Control Node</b>   | False zero reading from a dead gas sensor         | Not computed | Not computed | <b>4×2=8</b>  | Fail-safe hardware design where loss of sensor signal triggers an immediate system purge and lockout.        |
| <b>Outdoor Power Electronics &amp; Inverter Drive</b>     | Electrical fire originating in the inverter board | Not computed | Not computed | <b>4×2=8</b>  | Polyurethane potting of the PCB; strict isolation from the primary R290 mechanical bay.                      |
| <b>Outdoor Power Electronics &amp; Inverter Drive</b>     | Grid operator (DNO) rejection due to harmonics    | Not computed | Not computed | <b>3×2=6</b>  | Oversized dual AC line chokes and active power factor algorithms.                                            |
| <b>Indoor Hydronic Circulation &amp; Pressure Control</b> | System dry-running destroying the pump and BPHE   | Not computed | Not computed | <b>4×2=8</b>  | Dual redundancy of mechanical pressure switch and electronic flow sensor.                                    |
| <b>Indoor Hydronic Circulation &amp; Pressure Control</b> | Overpressurisation bursting hydraulic fittings    | Not computed | Not computed | <b>3×2=6</b>  | Calibrated PRV integrated strictly upstream of any isolator valves.                                          |

|                                                         |                                                           |              |              |               |                                                                                                 |
|---------------------------------------------------------|-----------------------------------------------------------|--------------|--------------|---------------|-------------------------------------------------------------------------------------------------|
| <b>Indoor Peak-Load &amp; Emergency Auxiliary Block</b> | Boiling internal fluid due to uncontrolled element firing | Not computed | Not computed | <b>5×1=5</b>  | Hardware-level redundancy with mechanical, non-resettable thermal fuse backups.                 |
| <b>Indoor Peak-Load &amp; Emergency Auxiliary Block</b> | Massive loss of overall building COP                      | Not computed | Not computed | <b>3×4=12</b> | Strict software lockouts preventing immersion use unless ambient drops below bivalent point.    |
| <b>System Telemetry &amp; Master Control Panel</b>      | Unintended operation loops freezing or boiling the system | Not computed | Not computed | <b>4×2=8</b>  | Hardware watchdog timer instantly resets processor on code hang; rigid parameter bounds.        |
| <b>System Telemetry &amp; Master Control Panel</b>      | Communication loss with commercial BMS                    | Not computed | Not computed | <b>2×3=6</b>  | Configurable fallback state requiring the unit to hold last known setpoint or target 45°C flow. |

### Data Sources

[LLM Synthesis] FMEA Matrix Generation

**Overall Section Grade: E**

# 7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

| Section                   | Grade | Primary Source Type            |
|---------------------------|-------|--------------------------------|
| 1. Brief & Requirements   | B     | User Input & Verified Search   |
| 2. Regulatory Posture     | D     | LLM Knowledge Base             |
| 3. Sizing Optimization    | B     | Deterministic Solver           |
| 4. System Modules & Specs | D     | Algorithmic Extraction + LLM   |
| 5. Economics Waterfall    | C     | Arithmetic Model + Estimations |
| 6. FMEA Risk              | E     | Pure LLM Synthesis             |

# 8. Audit Log

Pipeline execution trace for traceability and verification.

| #  | Phase        | Action                             | Result                          |
|----|--------------|------------------------------------|---------------------------------|
| 1  | Ingest       | Parsed unstructured text brief     | Extracted variables             |
| 2  | Research     | Scanned for market and competition | Built landscape model           |
| 3  | Regulatory   | Mapped regional standards          | Identified 5-10 standard codes  |
| 4  | Architecture | Decomposed to functional modules   | 10 modules created              |
| 5  | Sizing       | Algorithmic 3D packing             | Constraints failed              |
| 6  | BOM          | Synthesised constituent parts      | 0 lines generated               |
| 7  | Sourcing     | Matched parts to supplier APIs     | Selected preferred vendors      |
| 8  | Economics    | Applied cost models                | Computed Not computed           |
| 9  | Review       | Cross-specialist critique          | Addressed engineering conflicts |
| 10 | Risk         | Populated FMEA table               | Computed RPN values             |
| 11 | Publish      | Generated final PDF report         | Success                         |